Software Testing – 6 – Testing Strategies

Lecture # 28

1

Recap

- In the two last lectures, we had discussed black-box and white-box testing techniques in quite a bit of detail
- We had also discussed certain techniques within this these two broad testing approaches

Today's Lecture

- Today we'll talk about testing strategies

- Software testing needs to be planned in advance and conducted systematically
- A number of software testing strategies have been proposed in the literature
- All of them have these four generic characteristics

Characteristics of Testing Strategies - 1

- (1) Testing begins at the component level and works “outward” toward the integration of the entire computer-based system
- (2) Different testing techniques are appropriate at different points in time

Characteristics of Testing Strategies - 2

- (3) Testing is conducted by the developer of the software and (for large projects) an independent test group
- (4) Testing and debugging are different activities, but debugging must be accommodated in any testing strategy

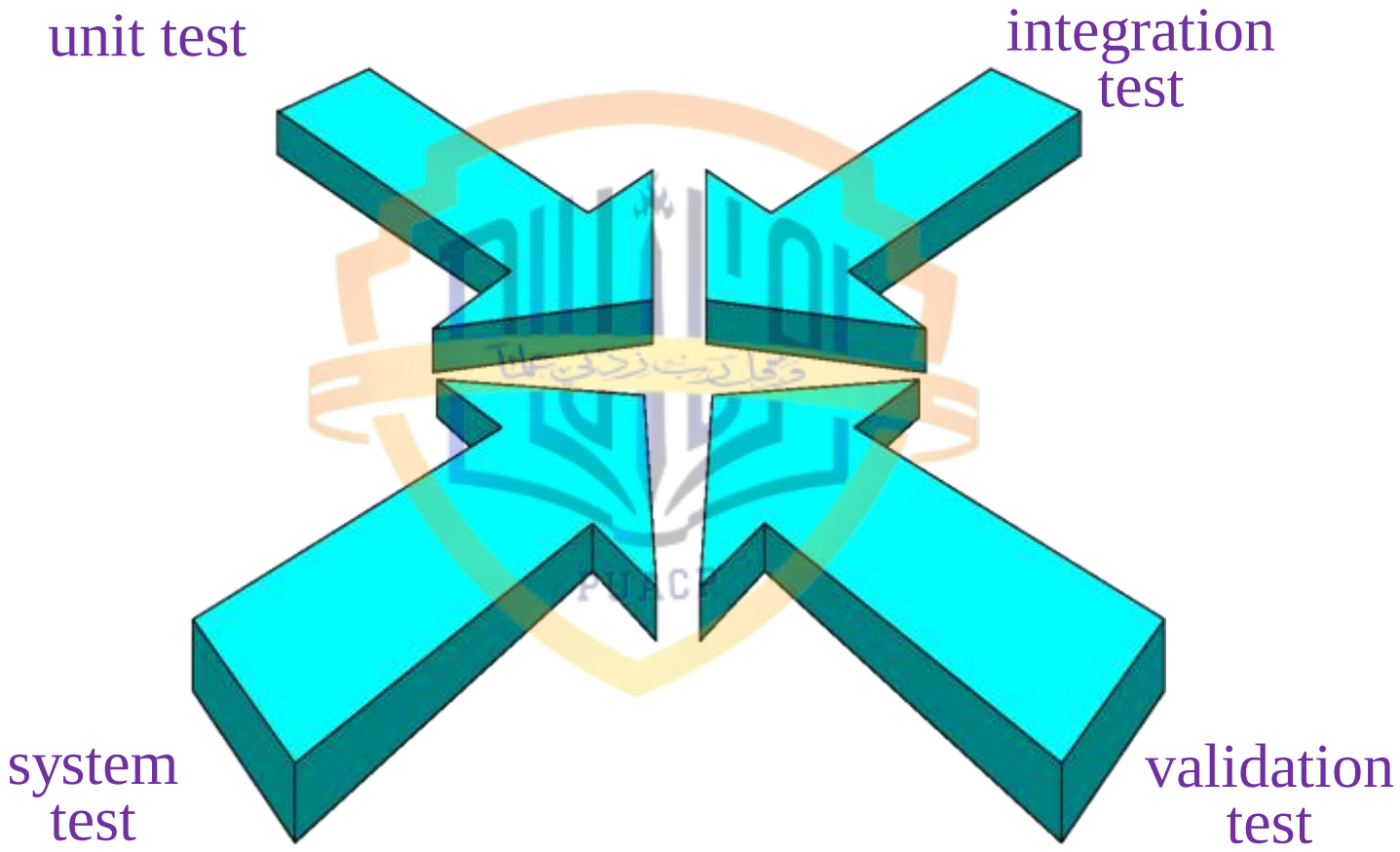
- A strategy for software testing must accommodate low-level tests that are necessary to verify that a small source code segment has been correctly implemented as well as high-level tests that validate major system functions against customer requirements

- A strategy must provide guidance for the practitioner and a set of milestones for the manager
- Because the steps of the test strategy occur at a time when deadline pressure begins to rise, progress must be measurable and problems must surface as early as possible

Testing Strategy - 1

- Unit testing
- Integration testing
- Validation testing
- System testing

Testing Strategy



Testing Strategy - 2

- Unit testing concentrates on each unit (i.e., component) of the software as implemented in source code
- Integration testing is next, where the focus is on design and the construction of the software architecture

Testing Strategy - 3

- Next is the validation testing, where requirements established as part of software requirements analysis are validated against the software that has been constructed
- Finally, we have system testing, where the software and other system elements are tested as a whole

Testing Strategy: Unit Testing

- Unit testing makes heavy use of white-box testing techniques, exercising specific paths in a module's control structure to ensure complete coverage and maximum error detection

Testing Strategy: Integration Testing

- Next, components must be assembled or integrated to form the complete software package
- Integration testing addresses the issues associated with the dual problems of verification and program construction
- BB test case design techniques are the most prevalent during integration

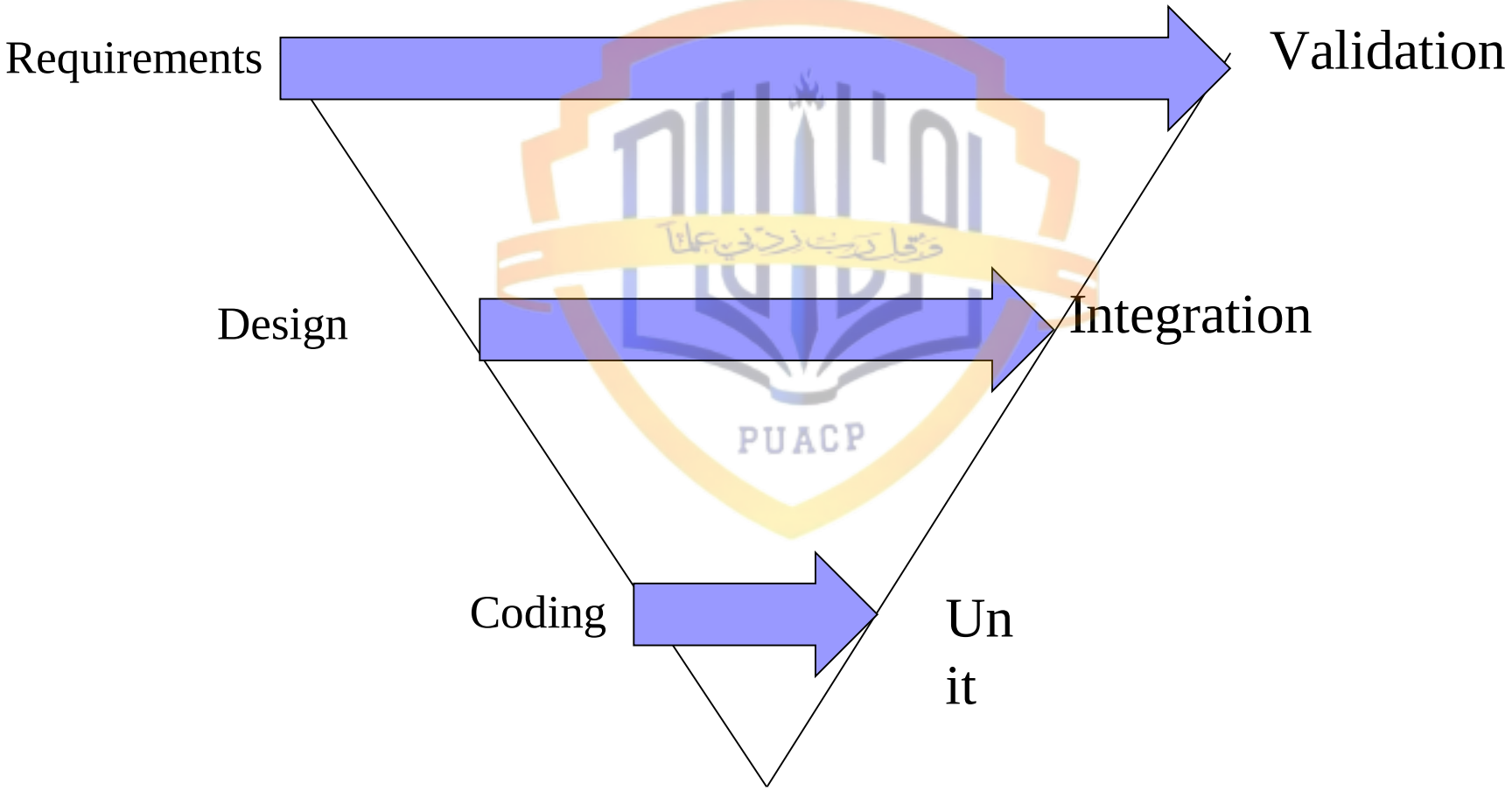
Testing Strategy: Validation Testing

- After software has been integrated, a set of high-order tests are conducted
- Validation criteria (established during requirements analysis) must be tested
- Validation testing provides final assurance that software meets all functional, behavioral, and performance requirements
- BBT techniques are used exclusively here

Testing Strategy: System Testing

- System testing is the last high-order testing and it falls outside the boundaries of software engineering and into the broader context of computer system engineering
- System testing verifies that all elements (hardware, people, databases) mesh properly and that overall system function/performance is achieved

The “V” Model of Software Testing



Strategic Issues - 1

- Specify product requirements in a quantifiable manner long before testing commences
  - NFRs
- State testing objectives explicitly
- Understand the users of the software and develop a profile for each user category
  - Actors/users of use-cases

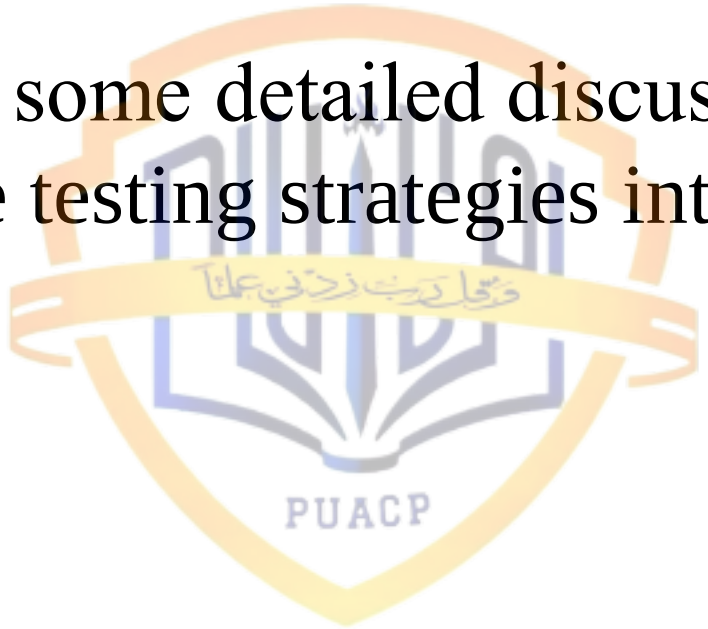
Strategic Issues - 2

- Develop a testing plan that emphasizes “rapid cycle testing”
- Build “robust” software that is designed to test itself
  - Software should be capable of diagnosing certain classes of errors
- Use effective formal technical reviews as a filter prior to testing

Strategic Issues - 3

- Conduct formal technical reviews to assess the test strategy and test cases themselves
- Develop a continuous improvement approach for the testing process

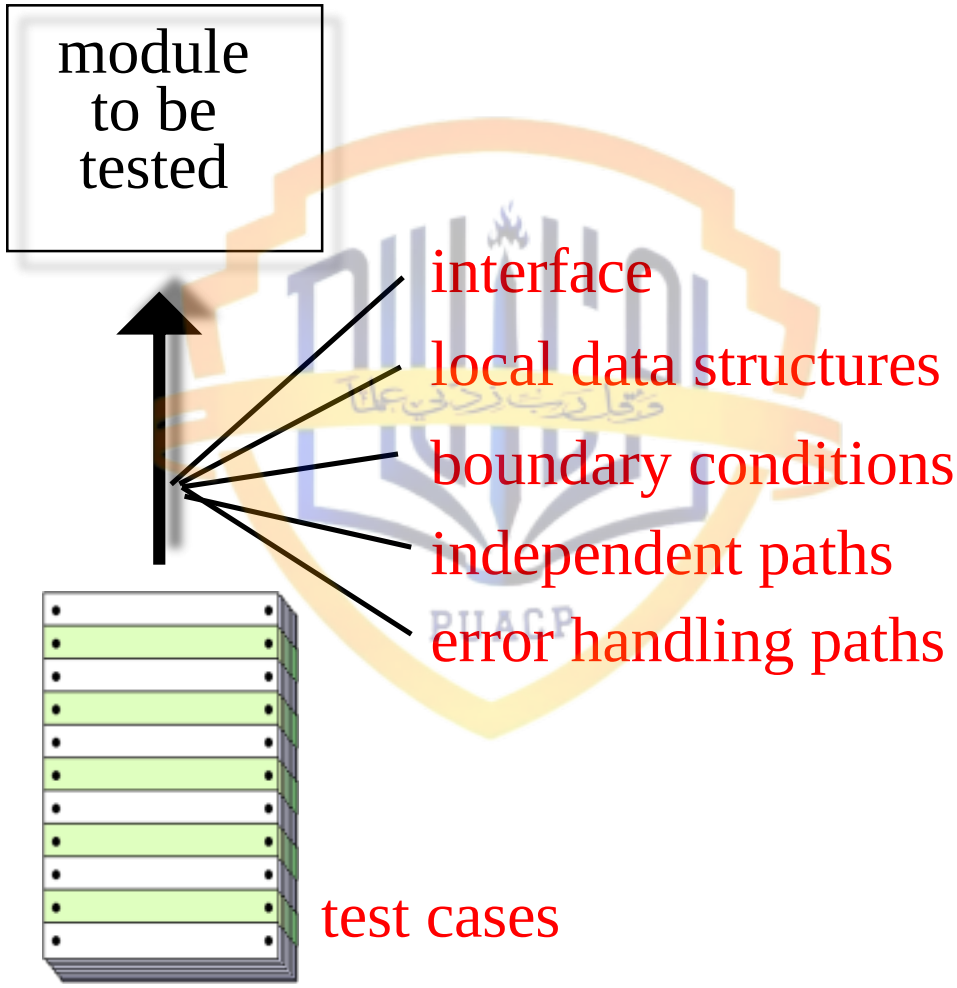
- Let’s have some detailed discussions on each of the testing strategies introduced



Unit Testing - 1

- Unit testing focuses verification effort on the smallest unit of software design – the software component or module
- The relative complexity of tests and uncovered errors is limited by the constrained scope established for unit testing
- Different units can be tested in parallel

Unit Testing



Unit Testing - 2

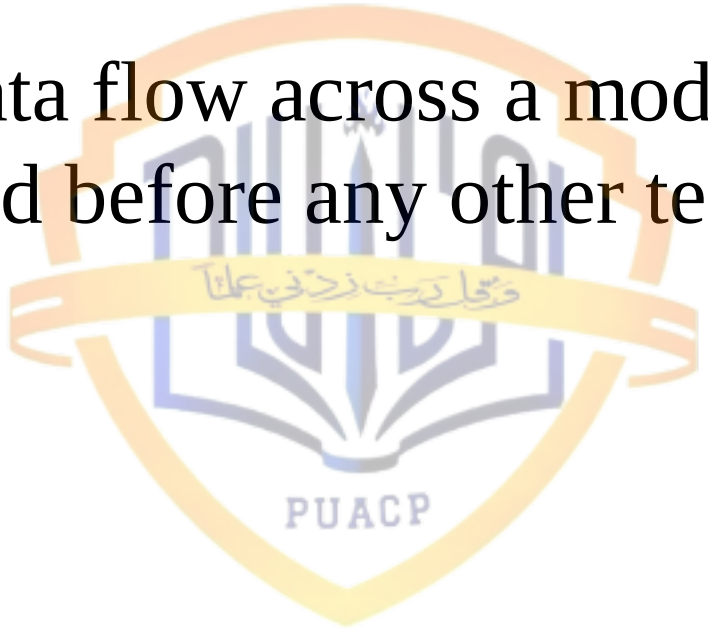
- The module interface is tested to ensure that information properly flows into and out of the program unit under test
- The local data structure is examined to ensure that data stored temporarily maintains its integrity during all steps in an algorithm's execution

Unit Testing - 3

- Boundary conditions are tested to ensure that the module operates properly at boundaries established to limit or restrict processing
- All independent paths through the control structure are exercised to ensure that all statements in a module have been executed
- All error handling paths are tested

Unit Testing - 4

- Tests of data flow across a module interface are required before any other test is initiated



Unit Testing: Common Errors Found

- Among the more common errors in computation are
  - Misunderstood or incorrect arithmetic precedence
  - Mixed mode operations
  - Incorrect initialization
  - Precision inaccuracy
  - Incorrect symbolic representation of expression

Unit Testing: Test Cases

- Test cases should uncover errors such as
  - Comparison of different data types
  - Incorrect logical operators or precedence
  - Expectation of equality when precision error makes equality unlikely
  - Incorrect comparison of variables
  - Improper or non-existent loop termination
  - Failure to exit when divergent iteration is encountered
  - Improperly modified loop variables

Unit Testing: Error Handling

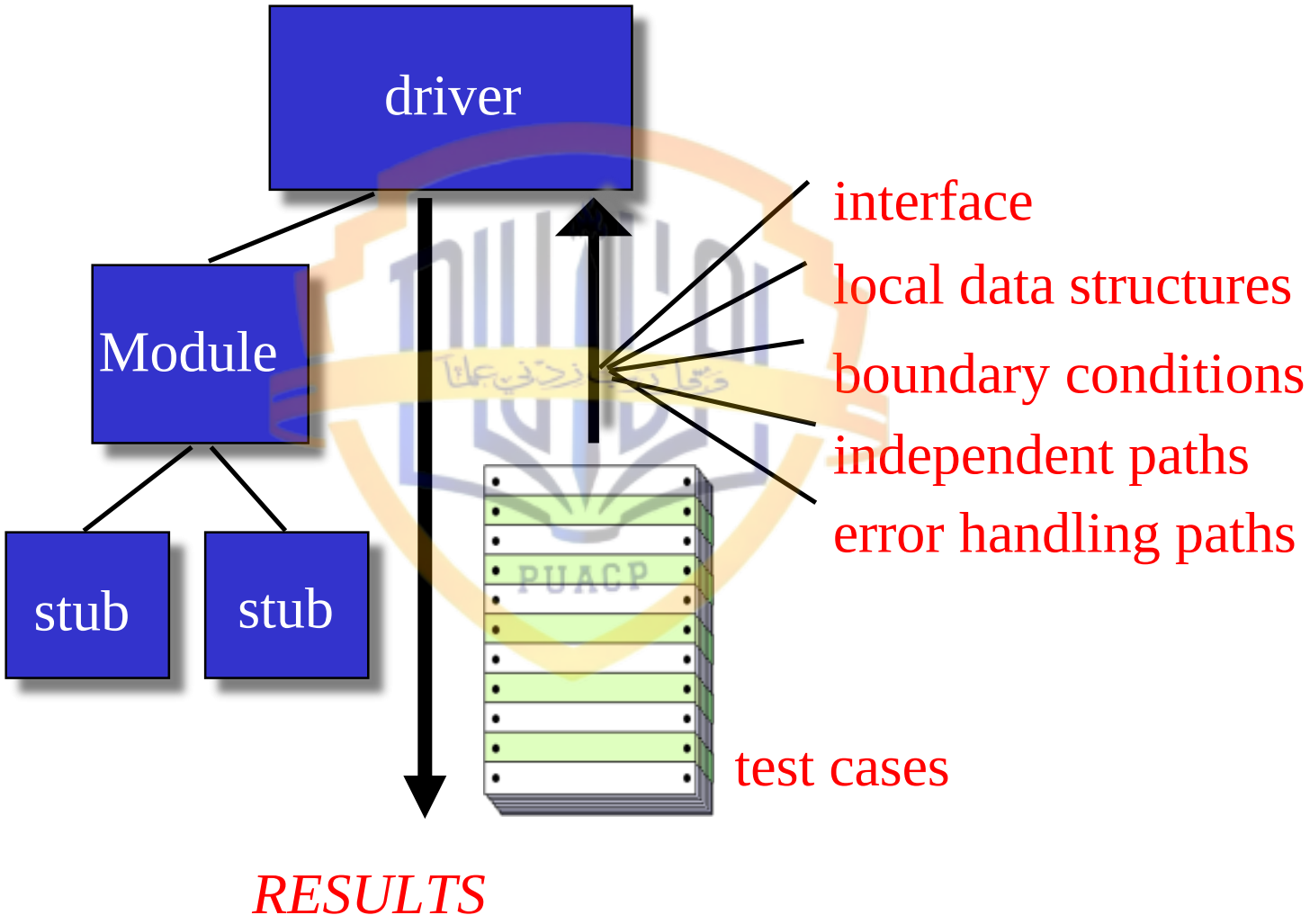
- Among the potential errors that should be tested when error handling is evaluated are
  - Error description is unintelligible
  - Error noted does not correspond to error encountered
  - Error condition causes system intervention prior to error handling
  - Exception-condition processing is incorrect
  - Error description does not provide enough information to assist in the location of the cause of the error

Unit Testing: Boundary Testing

- Boundary testing is the last and probably the most important task of unit testing step
- Software often fails at its boundaries
- Test cases that exercise data structure, control flow, and data values just below, at, and just above maxima and minima are very likely to uncover errors

- Unit testing is simplified when a component with high cohesion is designed
- When only one function is addressed by a component, the number of test cases is reduced and errors can be more easily predicted and uncovered
- Let’s see a unit test environment

Unit Test Environment



Integration Testing - 1

- "If they all work individually, why do you doubt that they'll work when we put them together?"
- The problem, of course, is "putting them together" – interfacing

Integration Testing - 2

- Data can be lost across an interface
- One module can have an inadvertent, adverse affect on another
- Sub-functions, when combined, may not produce the desired major function
- Individually acceptable imprecision may be magnified to unacceptable levels
- Global data structures can present problems

Software Integration

Integration Testing - 3

- Integration testing is a systematic technique for constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing
- Individual components have already been unit tested and the structure is dictated by design

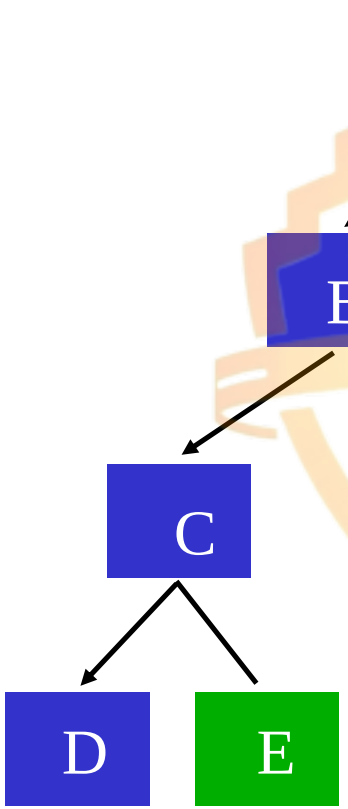
Integration Testing - 4

- There is tendency, sometimes, to attempt non-incremental integration – a sort of “big bang” approach
- All components are combined in advance
- The entire software is tested as a whole
- And chaos usually results!
- We should rather use incremental integration

Integration Testing Approaches

- Top-down integration
- Bottom-up integration
- Sandwich testing and integration

Top Down Integration



top module is tested with stubs

stubs are replaced one at a time, "depth first"

as new modules are integrated, some subset of tests is re-run

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Bottom-Up Integration

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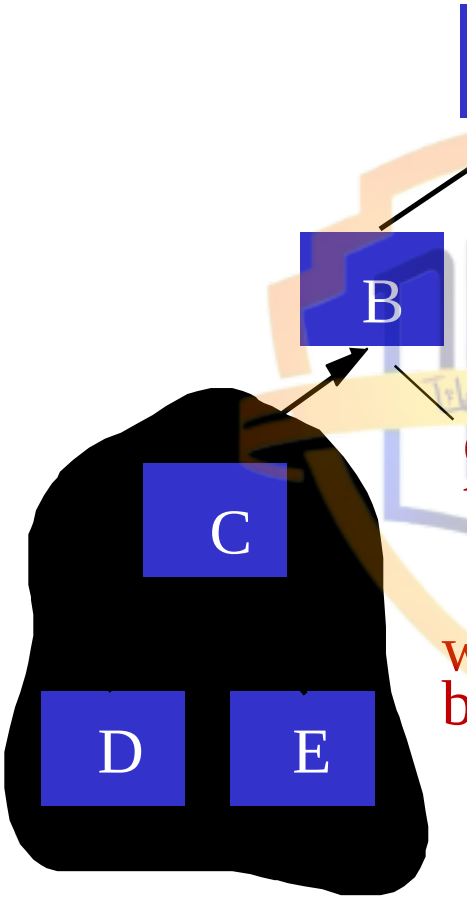
Bottom-Up Integration



drivers are replaced one at a time, "depth first"

worker modules are grouped into builds and integrated

cluster

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Sandwich Testing

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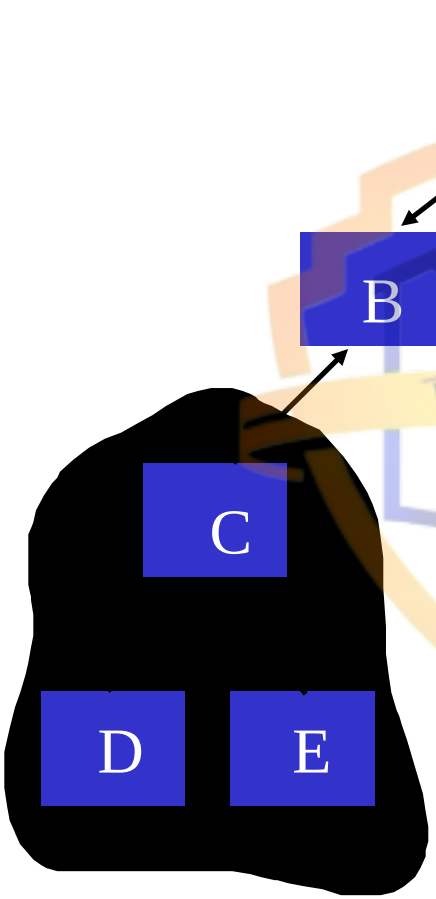
Sandwich Testing and Integration



Top modules are tested with stubs

Worker modules are grouped into builds and integrated

cluster

- Regression testing

Summary

- Today we've discussed different testing strategies

References

- Software Engineering: A Practitioner's Approach by Roger S. Pressman (Chapter 18.1 – 18.4)